

4.6.2 Discontinuous Function

A continuous function is a single unbroken curve. You could draw without lifting your pen from the paper. For example, here is a graph of a continuous function (Fig 4.31):



Fig. 4.31: Graph of Continuous Function

In case of a **discontinuous function**, you cannot draw on a graph without lifting your pencil. *There are three types of discontinuities:*

4.6.2.1 Asymptotic Discontinuities

Consider a function $f(x) = \frac{x+4}{(x-1)(x+8)}$

We know that its domain is limited to all real numbers except 1 and -8. Often, the most interesting points in a function are the problematic points, and indeed, we can see in the graph (Fig. 4.32) that the function behaves very strangely at the holes in the domain.

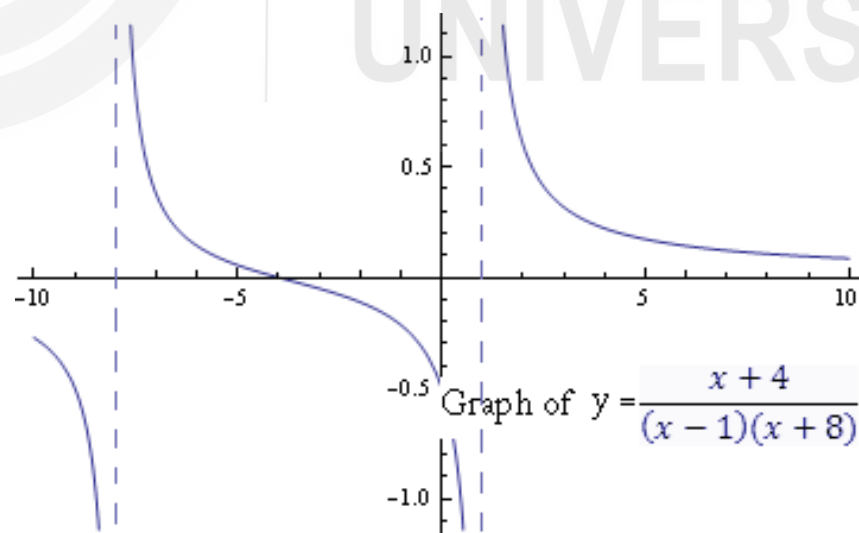


Fig. 4.32: Graph of Discontinuous Function

Look at the values (Table 4.6) of the function approaching one of the problematic points, $x=1$

Table 4.6: Values of $y=(x+4)/((x-1)(x+8))$

X	Y
0.25	-0.687
0.5	-1.059
0.75	-2.171
1.0	Undefined
1.25	2.270
1.5	1.158
1.75	0.786

As we approach $x = 1$ from the left side, with increasing x values, the y value drops to lower and lower negatives. The graph shows us that the y value seems to approach $-\infty$ from the left as the x -value approaches $x = 1$, marked by the dotted line. This represents a discontinuity, since the function is not connected over the dotted line. Specifically, this type of discontinuity is called an **asymptotic discontinuity**. The dotted lines represent **asymptotes**; they are values for which the function never takes a value, yet still approaches. The asymptotes we see with this function are called **vertical asymptotes** because they are vertical lines. There are also **horizontal asymptotes**. To see horizontal asymptotes, we graph the function $f(x)=2x/(x+4)$. It should have a horizontal asymptote at $y=2$.

4.6.2.2 Point Discontinuities

Consider the function

$$\begin{cases} 1, & \text{for } x=3 \\ f(x)=x^2, & \end{cases}$$

all other real values of x

Where as we have defined the value of the function to be 1 at $x=3$, and in the rest, the function is $f(x) = x^2$. As a result, the graph of the function is continuous except for the tiny hole in the curve at $x = 1$. It is discontinuous at a single point, viz. $x = 1$, and this discontinuity is called a **pointed is continuity**.

Point Discontinuities also arise when our function has a denominator that can be equal to zero, but that part of the denominator can also be cancelled out with a like term in the numerator. Consider the function $f(x)=x^2(x-2)/(x-2)$. If we try to find the value of the function at $x=2$, we get $f(2)=2^2(2-2)/(2-2) = 0/0$. As $0/0$ represents an undefined number, the function does not exist at that point.

However, if we restrict the function to a domain that does not include 2, we can simply cancel out the $(x-2)/(x-2)$ and be left with $f(x)=x^2$. This leaves us to define the function as:

$$f(x) = \begin{cases} x^2, & x \neq 2 \\ \text{undefined}, & x = 2 \end{cases}$$

The new definition has removed the discontinuity to show that the function behaves exactly like $f(x)=x^2$, except at $x=2$, where it is undefined.

4.6.2.3 Jump Discontinuities

$$\text{Consider the function } f(x) = \begin{cases} x^2, & x \leq 1 \\ 2-x, & x > 1 \end{cases}$$

For all intents and purposes, we can consider it two separate functions – one that is defined for x less than or equal to 1 and another function that is defined for x greater than 1. It is useful to think of it this way when we eventually use integration, differentiation, and other such mathematical tools. However, as it is written, f is a single function, called a **piecewise function**, since it is defined piece-by-piece. Note that the function adheres to our definition of being continuous (Fig. 4.33).

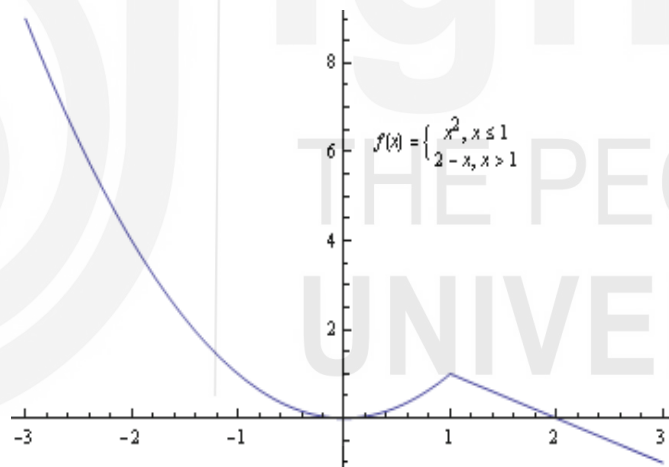


Fig. 4.33: Graph with Discontinuities

$$\text{Now, suppose we change our functions lightly to } f(x) = \begin{cases} x^2, & x \leq 1 \\ -x, & x > 1 \end{cases}$$

The two pieces now have a different value around $x = 1$, and we can see in the graph (Fig. 4.34) that our function f seems to "jump" from one branch to the other. Note this jump makes the function discontinuous. We refer to this as a **jump discontinuity**.

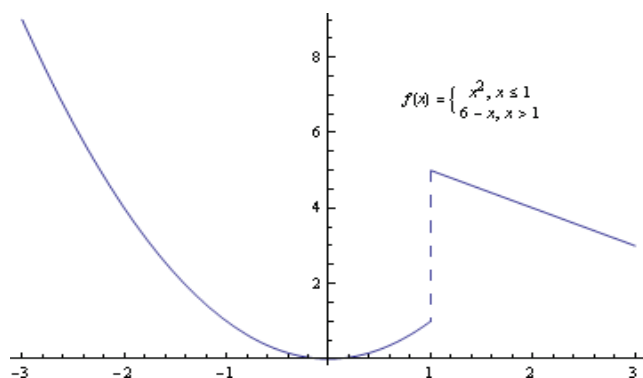


Fig. 4.34: Graph with Jump Discontinuities

Example:

We have to check the following function for discontinuities:

$$f(x) = \begin{cases} x+4/x, & x \leq 2 \\ x^3+1, & x > 2 \end{cases}$$

We begin by looking at the first branch of the function. First, note that it has an x in its denominator. This tells us that it has a problem in its domain at $x = 0$. As the values for x get very small, the value of the function approaches infinity, so we have an asymptotic discontinuity at $x = 0$. We also have to check if the branches correspond to the same value at $x=2$. The first branch yields $x+4 = 3$ and the second branch yields $2^3+1=9$. We can see the functions approach different values at $x=2$ so there is a jump discontinuity at $x=2$. There are no other discontinuities. (See Fig.4.35)

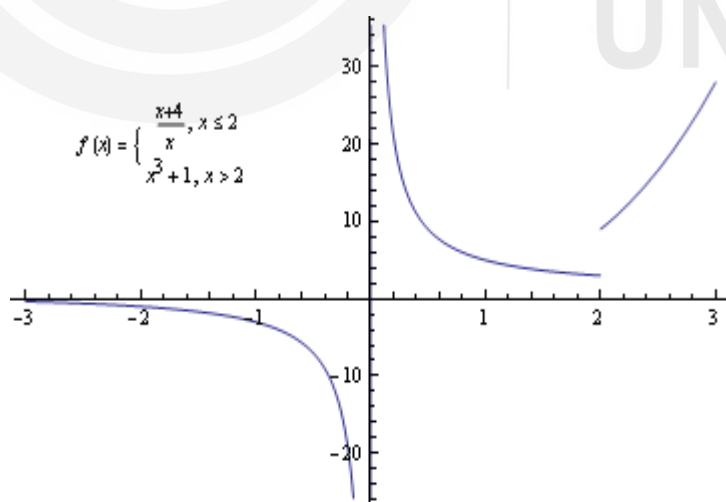


Fig. 4.35: Graph with Discontinuities

Check Your Progress 5

- 1) Draw the graph of $f(x) = -x^2 + 4$.

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- 2) State why $f(x) = x^3$ is also a continuous function. Draw its graph.

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- 3) Draw the graph of $f(x) = 1/x$. State why it is discontinuous.

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4.7 GRAPHING HYPERBOLA AND PARABOLA

We put together the functions that have characteristics of curves and pieces. In order to see such graphs, we need to see the intersection of a plane with a cone. Graphs of hyperbola and rectangular hyperbola which can be produced from such intersections are presented in the following.

4.7.1 Hyperbola

A hyperbola looks like two mirrored **parabolas**, with the two "halves" being called "branches". The definition of a *hyperbola* is the set of all points where the difference in the distance from two fixed points (called the *foci*) is constant.

There are two kinds of hyperbolas: horizontal and vertical.

The equation for a horizontal hyperbola is $(x-h)^2/a^2 - (y-b)^2/b^2 = 1$

The equation for a vertical hyperbola is $(y-v)^2/a^2 - (x-h)^2/b^2 = 1$

Notice that x and y switch places (as well as the h and v with them) to bring in the difference between horizontal and vertical hyperbolas.

The centre of a hyperbola is in between the two vertices of the hyperbola. While plotting, the centre is taken up first. A hyperbola has two axes of symmetry. The one that passes through the centre and the two foci is called the **transverse axis**; the one that is perpendicular to the transverse axis through the centre is called the **conjugate axis**. A horizontal hyperbola has its transverse axis at $y = v$ and its conjugate axis at $x = h$; a vertical hyperbola has its transverse axis at $x = h$ and its conjugate axis at $y = v$.

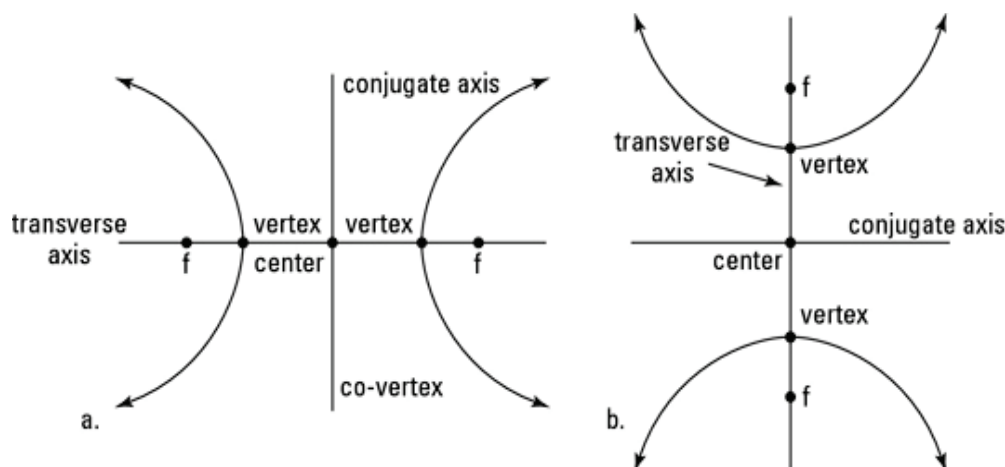


Fig. 4.36: Graph of Hyperbolas

You can see the two types of hyperbolas in Fig. 4.36: a horizontal hyperbola on the left, and a vertical one on the right. If the hyperbola that you are plotting is not in standard form, then you need to complete the square to get it in to standard form.

For example, the equation $\frac{(y-3)^2}{16} - \frac{(x+1)^2}{9} = 1$ is a vertical hyperbola. The centre (h,v) is $(-1,3)$. If $a^2 = 16$, $a = 4$, which means you count in vertical direction the number being under y variable given that $b^2 = 9$ or, $b = 3$. Thus, you count horizontally 3 units from the centre both to the left and to the right. The distance from the centre to the edge of the rectangle marked “ a ” determines half the length of the transverse axis, and the distance to the edge of the rectangle marked “ b ” determines the conjugate axis. In a hyperbola, a could be greater than, less than, or equal to b . If you count out a units from the centre along the transverse axis, and b units from the centre in both directions along the conjugate axis, these four points will be the midpoints of the sides of a very important rectangle. This rectangle has sides that are parallel to the x - and y -axis. This rectangle becomes a useful guide at the time of plotting the graph the hyperbola.

As may be seen from Fig. 4.36, hyperbolas contain other important parts. For example, a hyperbola has two vertices. There are two different equations—one for horizontal and one for vertically perbolas:

- A horizontal hyperbola has vertices at $(h \pm a, v)$.
- A vertical hyperbola has vertices at $(h, v \pm a)$.

The vertices for the above example are at $(-1, 3 \pm 4)$, or $(-1, 7)$ and $(-1, -1)$.

You find the foci of any hyperbola by using the equation $a^2 + b^2 = F^2$

where F is the distance from the centre to the foci along the transverse axis, the same axis that the vertices are on. The distance F moves in the same direction as a . As per the example under consideration, $16 + 9 = F^2$ or, $25 = F^2$

To name the foci as points in a horizontal hyperbola, you use $(h \pm F, v)$; to name them in a vertical hyperbola, you use $(h, v \pm F)$. The foci in the example would be $(-1, 3 \pm 5)$, or $(-1, 8)$ and $(-1, -2)$. Note that this places them inside the hyperbola.

Through the centre of the hyperbola run the asymptotes of the hyperbola. These asymptotes help in sketching the curves. Remember that the curves cannot cross asymptotes at any point on the graph.

The slopes of the asymptotes are $m = \pm a/b$ for vertical parabola or, $= \pm a/b$ for a horizontal parabola. To graph a hyperbola (Fig. 4.37), follow steps as given below:

- 1) Mark the centre: In our example hyperbola $(y-3)^2/16 - (x+1)^2/9 = 1$

You find that the centre of this hyperbola is $(-1, 3)$. Remember to switch the signs of the numbers inside the parentheses, and remember that h is inside the parentheses with, and v is inside the parentheses with y .

- 2) From the centre in Step 1, find the transverse and conjugate axes. Go up and down the transverse axis a distance of 4 (because 4 is under y), and then go right and left 3 (because 3 is under x). The points you marked as a (on the transverse axis) are your vertices.
- 3) Use these points to draw a rectangle that will help guide the shape of your hyperbola.
- 4) Draw diagonal lines through the centre and the corners of the rectangle that extend beyond the rectangle. This gives you two lines that will be your asymptotes.
- 5) Sketch the curves.

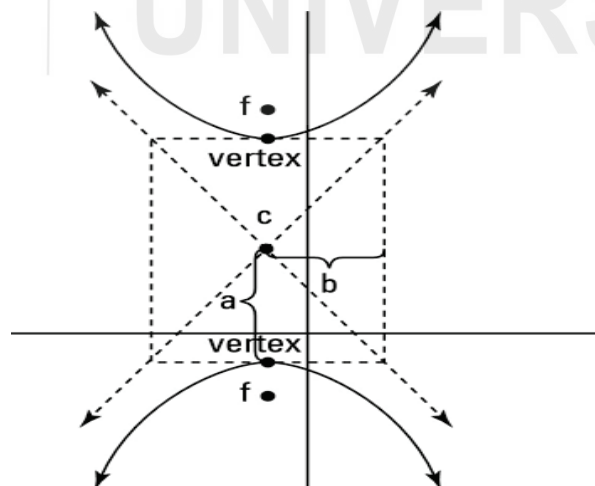


Fig 4.37: Graph of The Hyperbola

- 6) Draw the curves, beginning at each vertex separately, that hug the asymptotes the farther away from the vertices the curve gets.

Hyperbolas are used extensively in economics and finance (specifically port folio theory), where they can represent the various combinations of securities, funds,

etc., that yield similar risk-return ratios. Therefore, you often see efficient portfolio frontiers represented as partial hyperbolas. For similar reasons, production frontiers, which represent various combinations of capital and labor that produce a given output, as hyperbolas.

4.7.2 Difference between Hyperbola and Parabola

When a set of points present in a plane are equidistant from the directrix, a given straight line, and are equidistant from the focus, a given point which is fixed, it is called a parabola. When the difference of distances between a set of points present in a plane to two fixed foci or points is a positive constant, it is called a hyperbola.

All parabolas are of the same shape no matter what the size; all hyperbolas are of different shapes

The parabola is given by the equation $Y^2=X$; a hyperbola is given by the equation $XY=1$

In a parabola the two arms become parallel to each other where as in a hyperbola they do not.

4.7.3 Rectangular Hyperbola

It is a hyperbola with rectangular asymptotes. When the function is in the form $1/x^n$, the graph is a rectangular hyperbola. There are two types:

When $n = 2k + 1$, i.e., n is an odd integer, $y = 1/x$ (Fig. 4.38)

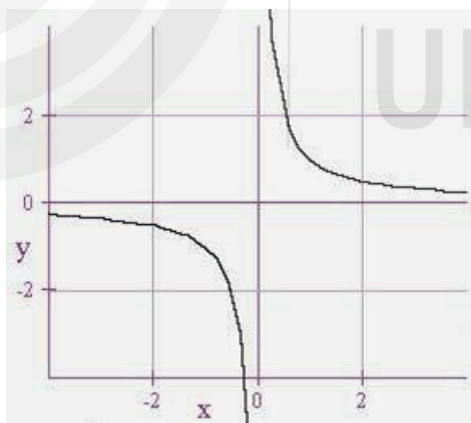


Fig 4.39: Rectangular Hyperbola $y=1/x^2$

Properties of the rectangular hyperbola:

- 1) There is a vertical asymptote at $x=0$.
- 2) There is a horizontal asymptote at $y=0$.
- 3) There are no stationary points.
- 4) There are no intercepts (since the axes are asymptotes. This is not true when the axes are not asymptotes.)

- 5) For $n = \text{odd integer}$, the gradient is always decreasing throughout the defined portion of x .
- 6) For $n = \text{even integer}$, the gradient when $x < 0$ is always increasing, whilst the gradient when $x > 0$ is always decreasing.

Example: Draw the graph of $y = \frac{x-1}{x+1}$

Ans. Intercepts: when $x = 0, y = -1$ and when $y = 0, x = 1$. Turning point: There are no turning points for this kind of graph. Asymptotes: As x tends to infinity, y tends to 1. As y tends to infinity, x tends to -1. $x = -1$ is a vertical asymptote and $y = 1$ is a horizontal asymptote. Gradient: The gradient is always increasing (Fig.4.40).

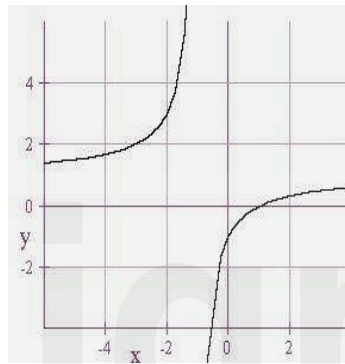


Fig 4.40: Graph of $y = \frac{x-1}{x+1}$

Rectangular hyperbola is used to represent unitary elasticity of demand, average fixed cost, transformation curve and indifference map to name a few examples.

Check Your Progress 6

- 1) Tell the difference between hyperbola and rectangular hyperbola.

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- 2) What would be the price elasticity of a rectangular hyperbola shaped demand?

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- 3) Find the centre of this hyperbola of the equation: $\frac{x^2}{81} - \frac{y^2}{9} = 1$

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4.8 GRAPHING FUNCTIONS WITH TWO VARIABLES

In the above discussion we have seen that for a function of one variable, $y=f(x)$, the graph of f is the set of all points (x,y) in 2-dimensional space such that $y=f(x)$. In general, these points lie on a curve in the plane. Now consider a function of two variables.

The graph of a function of two variables is a *surface in three dimensions*. Thus we deal with a **three-dimensional graph**, the graph of a function $z = f(x, y)$ of two variables, or the graph of a relationship $g(x, y, z)$ among three variables. For example, here (Fig. 4.41) is the graph of $z=2x^2+2y^2-4$.

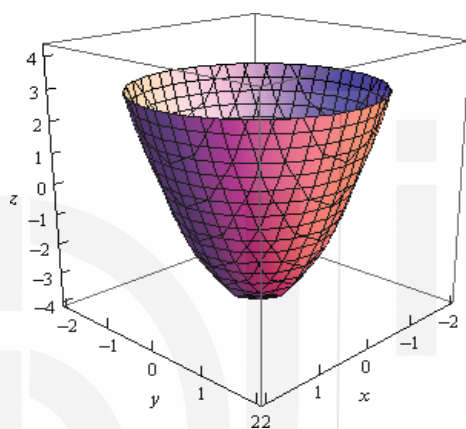


Fig. 4.41: Three-Dimensional Graph

The information in a three-dimensional graph is often collapsed into a two-dimensional graph with the use of contour lines. The procedure is: The x and y axes are retained, but instead of depicting z or $f(x,y)$ axis, all x, y combinations giving rise to the same z or $f(x, y)$ values are connected with a contour line; an arbitrary number of these may be shown for various values of z or $f(x,y)$.

4.8.1 Level Curves

Let f be a function of two variables, and c a constant. The set of pairs (x, y) such that $f(x, y)=c$ is called the **level curve** (or, contour lines) of f for the value c . **Example:** Let $f(x, y)=x^2 + y^2$ for all (x, y) . The level curve of f for value 1 is the set of all pairs (x, y) such that $x^2 + y^2=1$, a circle of radius 1. This set is shown in Fig. 4.42.

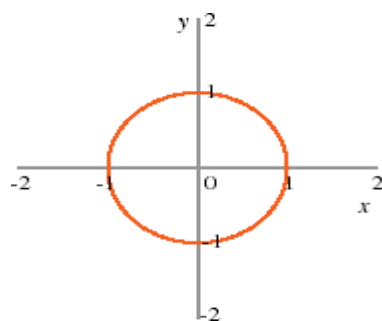


Fig. 4.42: Level curve